EEG Motor Imagery Classification

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Abstract

Brain–computer interfaces (BCIs) that decode motor imagery from electroencephalography (EEG) signals hold significant promise for assisting paralyzed individuals, but one limitation is that current approaches struggle to generalize across subjects. A model trained on one set of individuals' neural patterns often fails to effectively generalize to new users, limiting practical deployment. In this work, we systematically investigate how neural network architecture choices affect subject-independent generalization in EEG-based motor imagery classification. Using the BCI Competition IV Dataset 2a containing 9 subjects, 25 channels (22 EEG, 3 EOG), and 4 motor imagery classes (left-hand, right-hand, foot, and tongue) as our primary benchmark, we establish EEGNet8-2 (1) as a lightweight baseline and compare its cross-subject performance against several temporal modeling architectures. These include a hybrid CNN+LSTM and CNN+GRU models for spatial-temporal feature extraction, a Temporal Convolutional Network (TCN) for dilated temporal modeling, and a lightweight Transformer with spatial–temporal attention. We evaluate all models under subject-dependent tests against all 9 subjects and leave-one-subject-out (LOSO) protocol to directly measure generalization to unseen individuals and analyze how architectural components such as recurrent context, convolutional temporal modeling, and attention mechanisms contribute to robustness across subjects. Our results show that compact architectures like EEGNet (44.8% LOSO) and EEG-TCNet (45.2% LOSO) match or outperform larger recurrent and attention-based models, suggesting that strong inductive biases and moderate complexity are more important than architectural sophistication for cross-subject generalization.

1 Introduction

BCIs offer a direct communication pathway between the brain and external devices, providing a critical assistive technology for individuals with severe motor disabilities such as amyotrophic lateral sclerosis (ALS), spinal cord injury, and locked-in syndrome (2). Among non-invasive BCI paradigms, motor imagery (MI), the mental rehearsal of a movement without physical execution, has emerged as a widely studied approach. This is because it requires no external stimulus and can be performed volitionally by the user (3).

EEG data is the most common recording modality for MI-based BCIs due to its low cost and high portability. However, EEG signals are inherently noisy, non-stationary, and exhibit substantial inter-subject variability in both spatial topography and spectral characteristics (4). This variability poses a fundamental challenge. Models trained on data from a set of subjects frequently fail to generalize

to new, unseen individuals, necessitating lengthy per-user calibration sessions that limit practical deployment.

Recent work has explored deep learning architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and attention-based models for motor imagery classification, with compact CNN architectures such as EEGNet8-2 (1) commonly used as strong baselines.

In this work, we address this challenge by conducting a controlled comparison of five neural network architectures on the BCI Competition IV Dataset 2a (8) under both subject-dependent and LOSO cross-validation protocols. We investigate how specific architectural choices among recurrent context, bidirectionality, and attention-based temporal pooling all affect cross-subject robustness, with the goal of identifying which design principles most reliably improve subject-independent motor imagery decoding.

1.1 Background

The BCI Competition IV Dataset 2a is a widely used benchmark for 4-class motor imagery classification (With the 4 classes being left hand, right hand, both feet, tongue), recorded from 9 subjects across 22 EEG channels at 250 Hz. Each subject completed two sessions of 288 trials each, yielding 576 trials per subject.

Early approaches relied on hand-crafted spatial filters such as Common Spatial Patterns (CSP) combined with linear classifiers (5). While effective for a single subject, these methods depend on manually designed features that may not generalize across individuals. More recently, deep learning methods have shown competitive or superior performance while requiring minimal manual feature engineering. Schirrmeister et al. (6) introduced ShallowConvNet and DeepConvNet for end-to-end EEG decoding. Recurrent architectures such as LSTMs have been applied to capture temporal dependencies beyond the receptive field of standard CNNs, and Temporal Convolutional Networks (TCNs) (7) offer an alternative using dilated causal convolutions. More recently, Transformer-based models have been adapted for EEG, leveraging self-attention to capture long-range spatial-temporal relationships. Despite these advances, most evaluations focus on within-subject accuracy, and systematic comparisons of modern architectures under a strict subject-independent protocol remain limited.

2 Methods

2.1 Architectures

All architectures are evaluated under identical preprocessing and evaluation conditions to ensure a controlled comparison of architectural effects. Each model receives an input tensor of shape $(B, 22, 256)$ after performing a 128Hz resampling, representing multi-channel EEG trials and outputs logits of shape $(B, 4)$ corresponding to the four motor imagery classes.

2.1.1 EEGNet (Baseline)

Our baseline is EEGNet in its $(8, 2)$ configuration (1), a compact CNN ($\sim$2.6K parameters) that applies depthwise and separable convolutions to learn temporal and spatial filters directly from raw EEG data. It was specifically designed for BCI applications and has become a standard reference model for EEG classification tasks. The model accepts input of shape (B, C, T), which is unsqueezed to $(B, 1, C, T)$ internally, where B is the batch size, $C = 22$ EEG channels, and $T = 256$ timepoints. The architecture consists of two convolutional blocks followed by a linear classifier.

Block 1 performs temporal-then-spatial filtering. A temporal convolution with $F_1 = 8$ filters of kernel size $(1,\ 64)$ where $64 = \lfloor F_s/2 \rfloor$ at $F_s = 128$ Hz, capturing frequencies ≥ 2 Hz, is applied with same-padding and no bias, followed by batch normalization with no activation (linear, per the original paper). A depthwise convolution with kernel $(C,\ 1)$, depth multiplier $D = 2$, and `valid` padding then collapses the spatial channel dimension, producing $F_2 = F_1 \times D = 16$ feature maps (enforced, not a free parameter). This is followed by batch normalization, ELU activation, average pooling by a factor of 4, and dropout.

Block 2 refines features with a separable convolution: a depthwise temporal convolution with kernel $(1,\ 16)$ corresponding to 500 ms at the post-Block-1 effective rate of 32 Hz followed by a $(1,\ 1)$ pointwise convolution, batch normalization, ELU activation, average pooling by a factor of 8, and dropout. The flattened output feeds a single linear classifier with no hidden layers.

Max-norm constraints from the original paper are enforced after each optimiser step via a custom `MaxNormConstraint` that clips the ℓ_2 norm of parameter tensors in-place: $\|w\|_2 \leq 1.0$ on the depthwise convolution weights and $\|w\|_2 \leq 0.25$ on the classifier weights. Softmax is not applied in the forward pass; raw logits are returned and `nn.CrossEntropyLoss` (which subsumes log-softmax) is used during training. All batch normalization layers use `track_running_stats=False`, so normalization statistics are computed from the current mini-batch at both train and test time rather than from accumulated runningaverages. Under the LOSO protocol the test subject's signal distribution may differ substantially from the training population, and stale running statistics would introduce a systematic bias; recomputing statistics per batch avoids this at the cost of slightly higher variance in the normalised activations.

2.1.2 EEG-TCNet (Temporal Convolutional Network)

Unlike CNN-RNN hybrids, the TCN models long-range temporal dependencies entirely through convolutions with exponentially increasing receptive fields, avoiding recurrence altogether. The model accepts input of shape (B, C, T), which is unsqueezed to $(B, 1, C, T)$ internally, where $C = 22$ EEG channels and $T = 1000$ timepoints (4 s at 250 Hz).

The front-end mirrors only Block 1 of EEGNet. A temporal convolution with $F_1 = 8$ filters of kernel size $(1,\ 125)$ computed as $\lfloor 250/2 \rfloor$, capturing frequencies ≥ 2 Hz is applied with same-padding, followed by batch normalization and no activation (linear). A depthwise spatial convolution with depth multiplier $D = 2$ ($F_2 = F_1 \times D = 16$) collapses the channel dimension, followed by batch normalization, ELU, average pooling by a factor of 8 (reducing $T = 1000$ to 125 timesteps), and dropout ($p = 0.2$). Block 2's separable convolution is omitted, as the TCN blocks assume this role.

The TCN stage consists of 2 residual `TemporalBlock`s with exponentially increasing dilation factors $d \in \{1,\ 2\}$ and a fixed kernel size of 4. Each block contains two dilated causal convolutions: left-side padding of $(k - 1) \times d$ is applied and the corresponding right-side output is trimmed after each convolution to preserve sequence length and enforce causality. Each convolution is followed by batch normalization, ELU, and dropout ($p = 0.4$). A $1{\times}1$ projection handles the channel change from $F_2 = 16$ to `tcn_filters` $= 12$ in the first block; subsequent blocks use identity residuals. The TCN output is aggregated via global mean pooling across the 125 time steps, producing a 12-dimensional feature vector that passes through dropout ($p = 0.5$) and a linear classifier to produce logits for the four motor imagery classes.

2.1.3 CNN+LSTM

The CNN+LSTM replaces EEGNet's classifier with a multi-layer LSTM to model temporal dependencies across the CNN feature sequence. The model accepts input of shape (B, C, T), unsqueezed to $(B, 1, C, T)$ internally, where $C = 22$ EEG channels and $T = 256$ timepoints (2 s at 128 Hz). The CNN front-end uses $F_1 = 32$ temporal filters with kernel $(1,\ 64)$ computed as $\lfloor 128/2 \rfloor$, matching the EEGNet convention of capturing frequencies ≥ 2 Hz with same-padding of $(0,\ 32)$, followed by batch normalization (`track_running_stats=False`) and ELU, preserving the time dimension at 256 steps. A depthwise spatial convolution with `groups` $= F_1$ produces $F_2 = 64$ feature maps by collapsing the electrode dimension, followed by batch normalization, ELU, average pooling by a factor of 4 (reducing $T = 256$ to 64 timesteps), and dropout ($p = 0.5$). Block 2's separable convolution is omitted entirely, since the LSTM assumes this role.

The feature sequence is reshaped from $(B,\ F_2,\ 64)$ to $(B,\ 64,\ F_2)$ and fed into a 2-layer LSTM with 128 hidden units per layer and inter-layer dropout of 0.5. Rather than taking the final hidden state, temporal mean pooling is applied across all 64 LSTM output vectors, producing a 128-dimensional representation. This passes through dropout ($p = 0.5$) and a linear classifier. The same max-norm constraints as EEGNet are enforced after each optimiser step: $\|w\|_2 \leq 1.0$ on the depthwise convolution weights and $\|w\|_2 \leq 0.25$ on the classifier weights. At $\sim$200K trainable parameters, this is the largest model in our comparison, posing an increased overfitting risk on the relatively small LOSO training sets.

2.1.4 CNN+GRU with Temporal Attention

The CNN+GRU reuses the full EEGNet front-end (Block 1 and Block 2) but replaces the flatten-and-classify stage with a bidirectional GRU and learned attention pooling. The model accepts input of shape (B, C, T), unsqueezed to $(B, 1, C, T)$ internally, where $C = 22$ EEG channels and $T = 256$ timepoints (2 s at 128 Hz). The temporal convolution uses kernel size $(1,\ 64)$, computed as $\lfloor 128/2 \rfloor$, and the separable convolution in Block 2 uses kernel size $(1,\ 16)$, computed as $\lfloor 128/8 \rfloor$. After average pooling by 4 in Block 1 and by 8 in Block 2, the time dimension reduces from 256 to 8 steps. All batch normalization layers use `track_running_stats=False`, consistent with EEGNet, so statistics are recomputed from the current mini-batch at both train and test time.

The CNN output of shape $(B,\ T',\ F_2)$ with $T' = 8$ and $F_2 = 16$ is fed into a 2-layer bidirectional GRU with 48 hidden units per direction (96-dimensional output) and inter-layer dropout of 0.3. Layer normalization is applied to the full GRU output sequence for training stability. Instead of mean or last-step pooling, a soft temporal attention layer projects each GRU hidden state to a scalar score via a learned two-layer network ($\mathbb{R}^{96} \to \mathbb{R}^{48} \to \mathbb{R}^1$, with $\tanh$ nonlinearity), applies softmax to obtain weights α_t, and returns the weighted sum as the context vector. The attention weights are interpretable and can be visualized to identify which temporal frames the model deems most discriminative. The context vector passes through dropout ($p = 0.5$) and a linear classifier to produce logits for the four motor imagery classes. No max-norm constraints are applied, unlike EEGNet and the CNN+LSTM.

Hyperparameters were selected via a 3-round grid search optimizing mean validation accuracy across subjects A02, A03, and A07 (chosen as hard, strong, and mid-difficulty subjects). The final configuration uses CNN dropout of 0.4, GRU inter-layer dropout of 0.3, and classifier dropout of 0.5.

2.1.5 EEG Transformer with Temporal Self-Attention

To evaluate whether attention-based architectures improve motor imagery EEG classification, we implemented a lightweight Transformer model that combines the EEGNet-8,2 convolutional front-end with temporal self-attention. The model accepts input of shape (B, C, T), unsqueezed to $(B, 1, C, T)$ internally, where $C = 22$ EEG channels and $T = 256$ timepoints (2 s at 128 Hz). The convolutional front-end is identical to EEGNet Blocks 1 and 2 at 128 Hz with temporal kernel $(1,\ 64)$, separable kernel $(1,\ 16)$, pooling $\times 4$ then $\times 8$, reducing the time dimension from 256 to $T' = 8$ temporal tokens and producing a feature sequence of shape $(B,\ T',\ F_2)$ with $F_2 = 16$. The same max-norm constraints as EEGNet are enforced after each optimiser step: $\|w\|_2 \leq 1.0$ on the depthwise convolution weights and $\|w\|_2 \leq 0.25$ on the classifier weights. Note that the CNN batch normalization layers here use PyTorch's default `track_running_stats=True`, unlike the other models in this comparison.

The CNN features are linearly projected from $F_2 = 16$ to $d_{\text{model}} = 32$ and augmented with sinusoidal positional encodings (with dropout $p = 0.3$) before entering the transformer. Temporal dependencies are modeled using 2 Transformer encoder blocks, each with pre-norm layer normalization, multi-head self-attention (4 heads, 8 dimensions per head) with attention dropout, and a position-wise feedforward network ($d_{\text{ff}} = 64 = 2 \times d_{\text{model}}$, GELU activation, dropout $p = 0.3$) with residual connections. Because the CNN front-end compresses the signal to only 8 temporal tokens, the attention mechanism operates on a short sequence, keeping computational cost low. Attention weights of shape $(B,\ \text{heads},\ T',\ T')$ are extractable per layer, enabling post-hoc analysis of which temporal frames the model attends to. A final layer normalization is applied to the transformer output before aggregation.

The transformer outputs are aggregated using temporal mean pooling across the 8 tokens, passed through dropout ($p = 0.5$) and a linear classifier to predict the four motor imagery classes. The final model contains approximately $\sim$12K trainable parameters, making it substantially larger than EEGNet but still considerably smaller than the CNN+LSTM.

2.2 Data Preprocessing

Raw EEG signals were resampled from 250 Hz to 128 Hz, scaled to microvolts ($\times 10^6$), then bandpass filtered between 4 and 38 Hz using a 3rd-order Butterworth filter applied causally (`sosfilt`, single-pass forward only) to avoid any look-ahead artifacts at epoch boundaries. Only the 22 EEG channels were retained; the 3 EOG channels present in the dataset were discarded at trial extraction. Exponential running standardization (ERS) was applied to each channel on the continuous signal prior to trial

extraction, providing adaptive causal normalization without requiring a globally fitted normalizer, a property particularly relevant for the LOSO protocol where the test subject is entirely unseen during training.

Trials were extracted from 0.5 s to 2.5 s relative to cue onset (i.e. 2.5–4.5 s after trial onset), yielding segments of 256 time points per trial at 128 Hz. All models operate on this common input shape of $(B,\ 22,\ 256)$. Trials flagged as artifacts in the dataset metadata were excluded entirely; no additional artifact rejection was applied.

Data augmentation was applied during training: random temporal shifts of up to 64 samples with zero-padding (preserving input length), random amplitude scaling uniformly sampled from $[0.8,\ 1.2]$, and optional random sign flipping of entire trials.

2.2.1 Evaluation Protocol

We employ two evaluation protocols:

Subject-dependent. For within-subject evaluation, we use the Training session (T) split into training and validation sets (MI runs 3, 5, 6 for training; run 4 for validation), and the Evaluation session (E) as the held-out test set.

Leave-one-subject-out (LOSO). For subject-independent evaluation, we construct 36 folds (9 test subjects $\times$ 4 random repetitions). In each fold, 5 subjects' Training sessions serve as the training set, 3 subjects' Training sessions as the validation set, and the held-out subject's Evaluation session as the test set. Random assignment of subjects to train/validation roles is seeded for reproducibility. We report the mean and standard deviation of classification accuracy across all folds.

2.2.2 Training Details

All models were implemented in PyTorch and trained on NVIDIA GPUs using Google Cloud Platform (GCP) virtual machine instances. Training configurations were not fixed across architectures; instead, we explored a range of optimization and regularization settings for each model and benchmarked the configurations that produced the most stable and performant training behavior. Across experiments we evaluated variations of cross-entropy loss with label smoothing, Adam-based optimization with weight decay, and multiple learning rate schedules, including cosine annealing. Gradient clipping was applied in configurations where it improved training stability. For architectures derived from EEGNet, max-norm weight constraints from the original design (1.0 on depthwise convolution weights and 0.25 on classifier weights) were enforced after optimizer steps when those layers were present.

Early stopping strategies were also evaluated during experimentation. In the final benchmarking runs, early stopping was based on validation accuracy with a patience of 40 epochs using a 5-epoch smoothed validation metric, with a minimum of 100 training epochs before stopping could occur. Models were allowed to train for up to 300 epochs depending on convergence behavior. All batch normalization layers used `track_running_stats=False`, meaning normalization statistics were computed from the current mini-batch during both training and evaluation. Under the LOSO protocol the held-out subject's signal distribution may differ substantially from the training population, and relying on running statistics could introduce a systematic bias.

Batch size, dropout rates, learning rate schedules, and other regularization parameters were explored during development and selected empirically for the final benchmarking runs of each architecture.

3 Results

3.1 Overall Classification Accuracy

Table 1 summarizes the classification accuracy (4-class) for each architecture under both evaluation protocols.

EEGNet achieves the highest subject-dependent accuracy at 63.4%, and TCN leads under LOSO at 45.2%, though the differences between models are modest and within one standard deviation. Notably, the CNN+LSTM with $\sim$200K parameters (roughly 77$\times$ more parameters than EEGNet) achieves the second-lowest LOSO accuracy at 41.1%, suggesting that its larger capacity does not translate to better cross-subject generalization and likely overfits to training-subject patterns. The EEG Transformer,

Table 1: 4-class motor imagery classification accuracy (%) on BCI Competition IV Dataset 2a. Subject-dependent results report mean ± standard deviation across subjects. LOSO results report mean ± standard deviation across folds. Parameter counts correspond to the default configurations described in Section 2.1.

Model	Params	Subject-Dep. (%)	LOSO (%)
EEGNet (8,2)	∼2.6K	63.4 ± 16.7	44.8 ± 13.7
EEGNet-TCNet	∼3.8K	58.8 ± 16.8	45.2 ± 12.8
CNN+LSTM	∼200K	52.1 ± 16.1	41.1 ± 10.9
CNN+GRU (Attn)	∼68K	60.5 ± 17.3	44.7 ± 12.9
EEG Transformer	∼19K	55.3 ± 15.1	40.7 ± 12.2

despite incorporating temporal self-attention, achieves the lowest LOSO performance at 40.7%, performing comparably to the CNN+LSTM. In contrast, the compact EEGNet and TCN achieve the strongest LOSO results, suggesting that strong architectural inductive biases and limited model capacity are more important than expressive power for cross-subject EEG decoding.

The large standard deviations across all models (10–17% subject-dependent, 11–14% LOSO) indicate that inter-subject variability dominates over architectural differences, a well-documented challenge in EEG-based BCIs. This is further examined in the per-subject analysis below.

3.2 Per-Subject Analysis

Table 2 reports mean LOSO test accuracy per subject, averaged across the 4 repetitions per subject.

Table 2: Per-subject mean LOSO test accuracy (%), averaged across 4 repetitions. Subjects ordered by mean accuracy across models. Chance level is 25%.

Subject	EEGNet	CNN+GRU	TCN	CNN+LSTM	Transformer	Mean
A03	66.2	63.7	64.7	55.4	56.5	61.3
A01	57.6	57.1	58.8	52.3	58.5	56.9
A08	53.1	57.3	53.0	48.5	51.6	52.7
A09	56.0	48.8	52.0	48.9	43.8	49.9
A04	40.2	34.0	40.4	36.7	37.2	37.7
A07	35.3	34.1	**40.8**	30.1	32.6	34.6
A06	34.2	33.6	36.6	38.1	30.6	34.6
A05	33.1	34.1	34.6	29.5	26.5	31.6
A02	27.5	25.9	26.3	30.4	29.2	27.9

Table 3: Per-subject subject-dependent test accuracy (%) for all five architectures.

Subject	EEGNet	CNN+GRU	TCN	CNN+LSTM	Transformer
A01	72.6	76.5	64.8	49.5	64.1
A02	43.5	41.7	37.5	33.6	42.8
A03	83.5	84.6	77.3	68.9	70.0
A04	49.1	46.1	56.1	38.6	45.6
A05	39.9	41.3	34.1	32.6	33.3
A06	48.8	41.9	41.9	39.5	43.7
A07	73.3	59.2	61.0	59.6	47.3
A08	77.9	72.3	73.8	72.3	73.8
A09	82.2	81.1	83.0	74.6	77.3
Mean	63.4±16.7	60.5±17.3	58.8±16.8	52.1±16.1	55.3±15.1

The per-subject results reveal a consistent three-tier stratification of subjects that is largely independent of architecture. A03, A01, and A08 are high-performing subjects with mean LOSO accuracies above 50% across all models. A04, A07, A06, and A05 form a mid-difficulty tier (30–40%). Subject A02 is

a consistent outlier: no architecture exceeds 30.4% LOSO accuracy for this subject, barely above the 25% chance baseline. This pattern is consistent with prior work on this dataset (5), and likely reflects atypically weak or spatially diffuse mu/beta modulation in this individual rather than any architectural limitation.

Two subject-specific observations merit attention. First, on A07 under LOSO, TCN (40.8%) substantially outperforms all other architectures (30–35%), the largest single-subject advantage observed across the entire evaluation. A07 produces relatively consistent mu/beta signatures that are learnable cross-subject but require a sufficiently long temporal receptive field to capture, so TCN's dilated convolutions may be better matched to this subject's temporal dynamics than the GRU or LSTM's learned recurrence. Second, A09 shows the widest spread across architectures under LOSO (43.8% Transformer to 56.0% EEGNet), despite being among the easier subjects, suggesting that this subject's neural patterns are generalizable but architecture-sensitive in how they are extracted.

3.3 Frequency Band Ablation

Tables 4 and 5 report classification accuracy for each model when trained and evaluated on individual physiologically defined frequency bands under subject-dependent and LOSO protocols respectively.

Table 4: Subject-dependent 4-class accuracy (%, mean ± std across 9 subjects) by frequency band. **Bold** denotes the best result per model.

Band (Hz)	EEGNet	CNN+LSTM	CNN+GRU	TCN	Transformer
Delta (1–4)	45.3±10.9	45.0±9.0	55.2±10.3	50.5±5.5	43.7±9.4
Theta (4–8)	57.6±19.4	45.9±15.8	55.8±17.6	52.1±18.5	55.2±20.0
Mu (8–13)	61.4±19.4	50.8±19.4	57.7±20.4	55.7±18.7	57.6±21.2
Beta (13–30)	63.5±16.6	48.8±16.0	**68.4±15.5**	61.1±17.6	**62.4±19.6**
Gamma (30–40)	43.8±15.1	36.0±9.9	42.1±9.1	39.5±12.9	39.2±12.7
Mu+Beta (8–30)	67.0±13.5	**55.1±19.5**	67.3±16.2	61.7±15.5	60.9±21.5
Full (4–38)	**71.0±13.2**	55.1±19.2	67.1±16.6	**64.7±16.2**	60.8±22.3

Table 5: LOSO 4-class accuracy (%, mean ± std across 36 folds) for the mu+beta and full bands. **Bold** denotes the best result per model. Remaining bands were evaluated under the subject-dependent protocol only (Table 4).

Band (Hz)	EEGNet	CNN+LSTM	CNN+GRU	TCN	Transformer
Mu+Beta (8–30)	50.4±5.7	**46.8±5.0**	49.8±6.2	50.7±5.6	45.7±6.4
Full (4–38)	**50.7±6.2**	46.2±5.5	**50.3±6.3**	**51.1±6.1**	**45.9±6.0**

The results confirm that motor imagery classification is primarily driven by activity in the mu (8–13 Hz) and beta (13–30 Hz) bands, consistent with the known physiology of event-related desynchronization and synchronization over sensorimotor cortex (3). Across all five architectures, gamma (30–40 Hz) and delta (1–4 Hz) produce the lowest subject-dependent accuracies, with gamma falling near or below 44% for every model, which is only 19 percentage points above the 4-class chance level of 25%. The theta band (4–8 Hz) shows a modest improvement over delta but remains well below the mu and beta bands, indicating that slow cortical oscillations contribute little MI-discriminative signal under either evaluation protocol.

Beta consistently emerges as the single most informative band in the subject-dependent setting, outperforming mu by 2–11 percentage points across models. This asymmetry suggests that the post-cue beta event-related synchronization rebound is a more spatially consistent and subject-stable signature than the mu desynchronization during imagery itself, which is known to exhibit greater inter-subject variability in scalp topography.

Restricting to the combined mu+beta band incurs negligible degradation in LOSO performance, with losses of at most 0.5 percentage points relative to the full band for CNN+GRU, TCN, and EEGNet, and a marginal *improvement* of 0.6 pp for CNN+LSTM. This indicates that the frequency content outside 8–30 Hz contributes little generalizable information under cross-subject evaluation, and that the models are not relying on out-of-band artifacts to drive classification.

Two architectures show larger degradation from band restriction in the subject-dependent setting: EEGNet loses 4.0 pp moving from full to mu+beta, and TCN loses 3.1 pp. For EEGNet, this may reflect the depthwise spatial convolutions fitting subject-specific broadband signatures, which are patterns that are informative within a subject but do not transfer across individuals, as evidenced by the near-zero LOSO degradation (-0.3 pp) for the same model. For TCN, the dilated temporal convolutions with an exponentially growing receptive field may capture cross-band temporal co-modulation that is disrupted by narrow-band filtering, though again LOSO performance is largely unaffected (-0.4 pp). The delta band performance, while substantially below the full band, remains non-trivially above chance for CNN+GRU (55.2%) and TCN (50.5%), which may reflect residual low-frequency movement preparation signals or slow cortical potentials co-occurring with motor imagery.

Taken together, these ablations validate that the models' cross-subject generalization is grounded in neurophysiologically meaningful frequency content rather than broadband artifacts, supporting the interpretation of the LOSO results in Section 3.1.

4 Discussion

4.1 Insights

Model complexity does not predict cross-subject generalization. The CNN+LSTM ($\sim$200K parameters) achieves the lowest LOSO accuracy (41.1%), while EEGNet ($\sim$2.6K) and the EEG-TCNet ($\sim$3.8K) achieve the highest (44.8% and 45.2%). On the limited per-fold training data, larger models tend to overfit subject-specific patterns that do not transfer. The design of the CNN front-end appears to matter more than the choice of temporal back-end: the CNN+GRU reuses EEGNet's compact spatial filters and matches its LOSO performance, whereas the CNN+LSTM's wider front-end ($F_1{=}32$, $F_2{=}64$) achieves lower LOSO accuracy despite extracting richer within-subject features. Neither recurrence nor attention provides a measurable LOSO improvement over EEGNet's fixed pooling.

Across all architectures, inter-subject variability dominates over architectural differences. A consistent three-tier subject stratification emerges, with subject A02 remaining near chance for every model, indicating that for certain individuals, cross-subject transfer faces limits that architecture alone cannot address.

4.2 Best Approaches

The EEG-TCNet achieves the best LOSO accuracy (45.2%) with only $\sim$3.8K parameters, closely followed by EEGNet (44.8%) and the CNN+GRU (44.7%). For within-subject decoding, EEGNet leads at 63.4% and offers the best accuracy-to-parameter ratio overall. The CNN+GRU's temporal attention weights provide an interpretability advantage, though this does not translate to improved accuracy. Restricting input to the mu+beta band (8–30 Hz) incurs at most 0.5 pp of LOSO degradation, making it a practical choice when artifact rejection or efficiency is a priority.

4.3 Why This Was the Case

EEG signals are known to vary substantially across individuals in amplitude, spatial topography, and spectral profile (4), and this inter-subject variability is the dominant factor in our results. The consistent subject stratification across all five architectures as well as the large gap between subject-dependent and LOSO accuracy for every model confirm that differences between subjects outweigh differences between architectures. With 9 subjects and only 5 training subjects per LOSO fold, models with more free parameters have greater capacity to memorize subject-specific patterns in the training set, which explains why the CNN+LSTM ($\sim$200K parameters) generalizes worst despite reasonable within-subject performance.

The frequency band ablation provides additional grounding: all models derive their cross-subject discriminative signal from the mu and beta bands, consistent with the established physiology of sensorimotor event-related desynchronization (3). The negligible LOSO degradation when restricting to 8–30 Hz confirms that no model relies on out-of-band artifacts, and that the performance differences across architectures reflect how effectively each model extracts and generalizes from this shared neurophysiological signal rather than exploiting dataset-specific confounds.

4.4 Limitations and Future Work

This study is constrained by its small subject pool (9 subjects), reliance on a single dataset, lack of domain adaptation or fine-tuning, and uneven hyperparameter tuning across architectures. Validating on larger benchmarks such as the PhysioNet EEG Motor Movement/Imagery Dataset would strengthen generality. Exploring methods that directly address inter-subject variability at the data level such as spatial alignment or domain adaptation is a natural next step, as our results suggest this variability is the primary barrier to cross-subject transfer. Extending comparisons to conformers and state-space models is also warranted.

5 Conclusion

In this work, we systematically compared five neural network architectures for subject-dependent and leave-one-subject-out EEG motor imagery classification on the BCI Competition IV Dataset 2a. Compact architectures with strong domain-specific inductive biases such as EEGNet (44.8% LOSO) and EEG-TCNet (45.2% LOSO) match or outperform substantially larger recurrent and attention-based models, including a 200K-parameter CNN+LSTM (41.1% LOSO) and an EEG Transformer (40.7% LOSO). This inversion of the capacity-performance relationship under cross-subject evaluation is a central finding: model complexity does not predict generalization, and may actively harm it when training data is limited to a small number of subjects.

Per-subject analysis reveals that inter-subject variability dominates over architectural differences, with a consistent three-tier subject stratification observed across all models. Subject A02 remains near chance for every architecture, suggesting a fundamental limit on cross-subject transfer for certain individuals that no purely architectural intervention can overcome.

Frequency band ablations confirm that cross-subject generalization is grounded in neurophysiologically meaningful signal: restricting to the mu+beta band (8–30 Hz) incurs at most 0.5 percentage points of LOSO degradation across all five models, validating that the models rely on motor-imagery-relevant oscillatory activity rather than broadband artifacts.

Taken together, these results suggest that architectural inductive biases aligned with EEG signal structure, such as depthwise spatial convolutions and dilated temporal convolutions, are more important than expressive capacity for generalizable motor imagery decoding, and that future progress may depend more on addressing spatial mismatch across subjects than on temporal modeling sophistication.

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